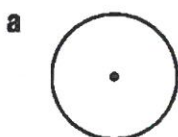


Name: _____

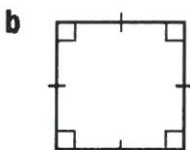
Measurement & Area

(4 Marks)

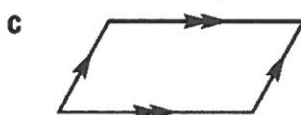
1 Name these shapes.



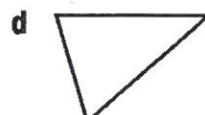
circle



square



parallelogram



Tri

(6 Marks)

2 Write the missing number.

a 1 km = 1000 m

b 1 m = 100 cm

c 1 cm = 10 mm

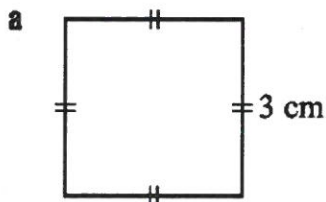
d 1 L = 1000 mL

e 0.5 km = 500 m

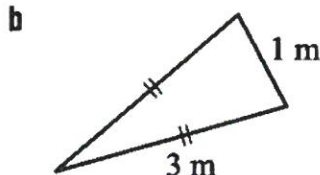
f 2.5 cm = 25 mm

(3 Marks)

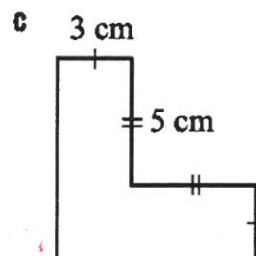
3 Find the perimeter of these shapes.



$3 \times 4 = 12 \text{ cm}$



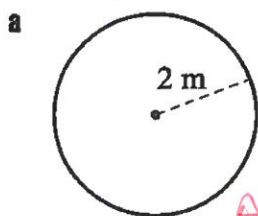
$3 + 3 + 1 = 7 \text{ m}$



$8 + 8 + 5 + 5 + 3 + 3 = 32 \text{ cm}$

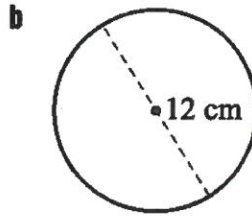
(4 Marks)

4 Use these rules to find the circumference (C) and area (A) of the circles correct to one decimal place: $C = 2\pi r$ and $A = \pi r^2$ where r is the radius.



$C = 2 \times \pi \times r$
 $= 2 \times \pi \times 2$
 $= 12.57 \text{ m}$

$A = \pi r^2$
 $= \pi (2)^2$
 $= 12.57 \text{ m}^2$



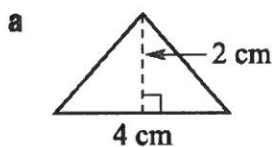
$C = 2\pi r$
 $= 2 \times \pi \times 6$
 $= 37.70 \text{ cm}$

$A = \pi r^2$
 $= \pi (6)^2$
 $= 113.10 \text{ cm}^2$

(6 Marks)

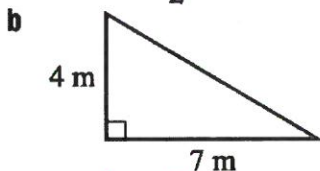
- 5 Find the area of these triangles using $A = \frac{1}{2}bh$.

$$A = \frac{1}{2}b \times h$$



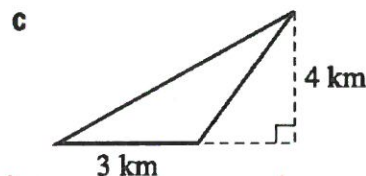
$$A = 0.5 \times 4 \times 2$$

$$= 4 \text{ cm}^2$$



$$A = 0.5 \times 4 \times 7$$

$$= 14 \text{ m}^2$$



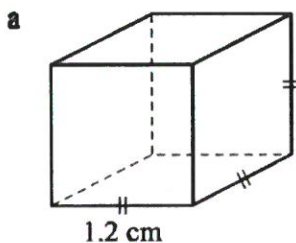
$$A = 0.5 \times 3 \times 4$$

$$= 6 \text{ km}^2$$

Surface Area

(9 Marks)

- 6 Find the total surface area of these solids.

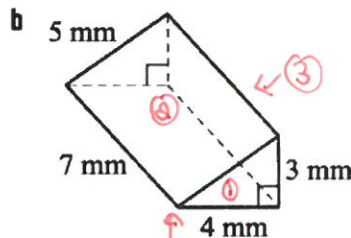


$$A = L \times W$$

$$= 1.2 \times 1.2$$

$$= 1.44 \text{ cm}^2 \times 6$$

$$= 8.64 \text{ cm}^2$$



$$A_1 = \frac{1}{2}b \times h$$

$$= \frac{1}{2}(7 \times 4)$$

$$= 14 \text{ mm}^2 \times 2$$

$$= 28 \text{ mm}^2$$

$$A_2 = L \times W$$

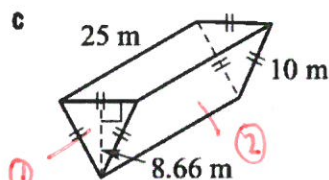
$$= 5 \times 7$$

$$= 35 \text{ mm}^2$$

$$A_3 = 3 \times 7$$

$$= 21 \text{ mm}^2$$

$$TSA = 91 \text{ mm}^2$$



$$A_1 = \frac{1}{2}b \times h$$

$$= \frac{1}{2}(25 \times 8.66)$$

$$= 108.125 \times 2$$

$$= 216.25 \text{ m}^2$$

$$A_2 = L \times W$$

$$= 25 \times 10$$

$$= 250 \text{ m}^2 \times 3$$

$$= 750 \text{ m}^2$$

$$TSA = 966.25 \text{ m}^2$$

- 7 Use Pythagoras' theorem to determine any unknown side lengths and find the total surface area of these solids, correct to one decimal place.

a

$$a^2 + b^2 = c^2$$

$$2^2 + 3^2 = c^2$$

$$c^2 = 13$$

$$c = 3.61$$

$$A_1 = \frac{1}{2}b \times h$$

$$= \frac{1}{2}(2 \times 3)$$

$$= 3 \times 2 = 6$$

$$A_2 = L \times W$$

$$= 2 \times 7$$

$$= 14$$

$$A_3 = L \times W$$

$$= 3.61 \times 7$$

$$= 25.27$$

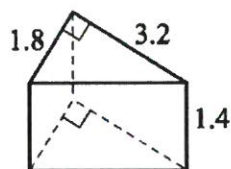
$$TSA = 66.24 \text{ units}^2$$

b

$$a^2 + b^2 = c^2$$

$$(1.8)^2 + (3.2)^2 = c^2$$

$$c = 3.67$$



$$A_1 = \frac{1}{2}b \times h$$

$$= \frac{1}{2}(1.8 \times 3.2)$$

$$= 2.88 \times 2 = 5.76$$

$$A_2 = 3.2 \times 1.4$$

$$= 4.48$$

$$A_3 = 1.8 \times 1.4$$

$$= 2.52$$

$$A_4 = 3.67 \times 1.4$$

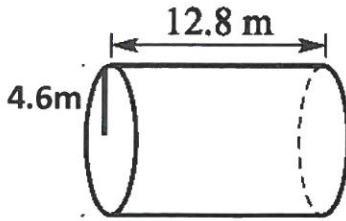
$$= 5.14$$

$$TSA = 17.9 \text{ unit}^2$$

(8 Marks)

8 Find the total surface area of these solids.

a



$$C = 2\pi r$$

$$= 2\pi(4.6)$$

$$= 28.90\text{ m}^2$$

$$A_1 = \pi r^2$$

$$= \pi(4.6)^2$$

$$= 66.48\text{ m}^2$$

$$A_2 = L \times W$$

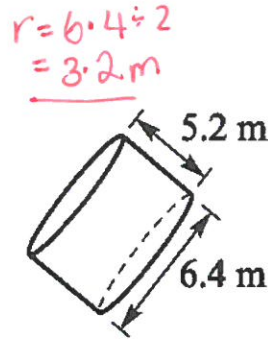
$$= 12.8 \times 28.90$$

$$= 369.95\text{ m}^2$$

$$\times 2 = 132.96\text{ m}^2$$

$$\text{TSA} = 502.91\text{ m}^2 \checkmark$$

b



$$r = 6.4 \div 2$$

$$= 3.2\text{ m}$$

$$C = 2\pi r$$

$$= 2\pi \times 3.2$$

$$= 20.11\text{ m}$$

$$A = \pi r^2$$

$$= \pi \times (3.2)^2$$

$$= 32.17\text{ m}^2$$

$$A = L \times W$$

$$= 5.2 \times 20.11$$

$$= 104.57\text{ m}^2$$

$$\times 2$$

$$64.34\text{ m}^2$$

$$\text{TSA} = 168.89\text{ m}^2 \checkmark$$

Volume

9. Find the volume of a cube of side 15 m.

(2 Marks)

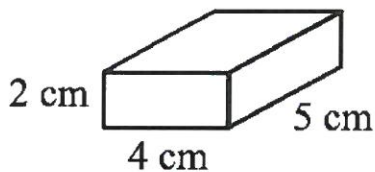
$$V = L \times W \times H$$

$$= 15 \times 15 \times 15 = 3375\text{ m}^3$$

(8 Marks)

10. Calculate the volume of these 3D figures

a)

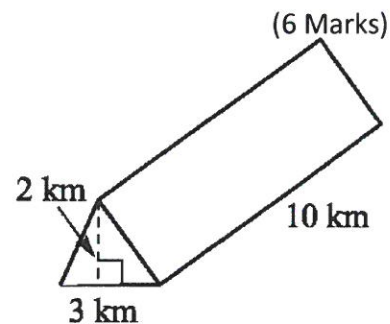


$$V = L \times W \times H$$

$$= 2 \times 4 \times 5$$

$$= 40\text{ cm}^3$$

b)



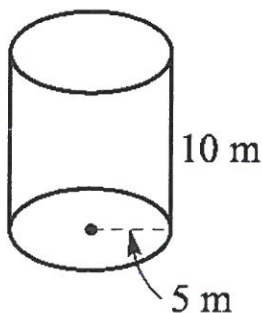
$$V = \frac{1}{2} b \times h \times l$$

$$= \frac{1}{2} (2 \times 3)$$

$$= 3 \times 10$$

$$= 30\text{ km}^3$$

c)



$$V = \pi r^2 \times h$$

$$= \pi (5)^2 \times 10$$

$$= 785.40\text{ m}^3$$

Extension:

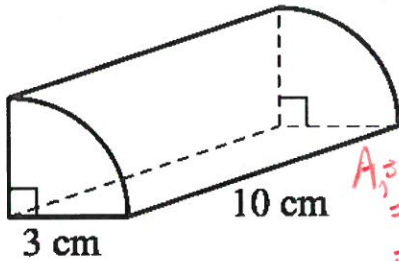
(4 Marks)

Calculate the surface area and Volume of the following

(6 Marks)

16

a



$$A_1 = L \times W$$

$$= 10 \times 3$$

$$= 30 \times 2$$

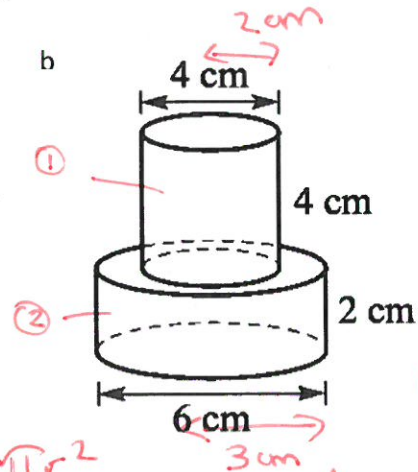
$$= 60 \text{ cm}^2$$

$$TSA = 121.25 \text{ cm}^2 \checkmark$$

$$V = 7.07 \times 10$$

$$= 70.7 \text{ cm}^3$$

b



$$A_1 = \pi r^2$$

$$= \pi (2)^2$$

$$= 12.57 \text{ cm}^2$$

$$A_2 = 12.57 \times 4$$

$$= 50.27 \text{ cm}^2$$

$$SA_1 = 62.84 \text{ cm}^2$$

$$C = 2\pi r$$

$$= 2\pi \times 2$$

$$= 12.57 \text{ cm}$$

$$C = 2\pi r$$

$$= 2\pi \times 3$$

$$= 18.85 \text{ cm}$$

$$A_2 = \pi r^2$$

$$= \pi (3)^2$$

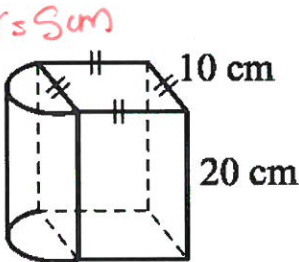
$$= 28.27 \text{ cm}^2$$

$$A_{\text{top}} = 28.27 - 12.57$$

$$= 15.70 \text{ cm}^2$$

(6 Marks)

c



$$A = L \times W$$

$$= 10 \times 10$$

$$= 100 \text{ cm}^2 \times 2$$

$$A = L \times W = 200 \text{ cm}^2$$

$$= 10 \times 20$$

$$= 200 \times 3$$

$$= 600 \text{ cm}^2$$

$$SA = 800 \text{ cm}^2$$

$$V = L \times W \times h$$

$$= 10 \times 10 \times 20$$

$$= 2000 \text{ cm}^3$$

$$C = 2\pi r$$

$$= 2 \times \pi \times 5$$

$$= 31.42 \text{ cm}$$

$$A_1 = \pi r^2$$

$$= \pi (5)^2$$

$$= 78.54 \text{ cm}^2 \div 2$$

$$39.27 \text{ cm}^2 \times 2$$

$$78.54 \text{ cm}^2$$

$$A_2 = L \times W$$

$$= 31.42 \times 20$$

$$= 428.4 \text{ cm}^2 \div 2$$

$$SA = 214.26 \text{ cm}^2$$

$$292.74 \text{ cm}^2$$

$$V = 39.27 \times 20$$

$$= 785.40 \text{ cm}^3$$

$$TSA = 144.51 \text{ cm}^2$$

$$= 2 \times 18.85$$

$$= 37.70 \text{ cm}^2$$

$$V_1 = 12.57 \times 4$$

$$= 50.28 \text{ cm}^3$$

$$V_2 = 28.27 \times 2$$

$$= 56.54 \text{ cm}^3$$

$$ISA = 81.67 \text{ cm}^2$$

$$\text{Total Volume} = 106.82 \text{ cm}^3$$

$$\text{Total SA} = 1092.70 \text{ cm}^2$$

$$\text{Total Volume} = 2785.40 \text{ cm}^3$$